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Floor rolling apparatus

Abstract

Embodiments of the present invention include a floor rolling apparatus including a pair of longitudinal support members with a respective pair of upper vaulted ends, an upper lateral support connecting from one longitudinal support member to the other, a lateral grip connecting from one longitudinal support member to the other, a lower reinforcement assembly, and a roller assembly with removably attached roller. The longitudinal support members include a first leveraging section and a second leveraging section that together with the upper lateral support are structured to leverage a user's weight, lower arms and upper arms to facilitate rolling the apparatus roller on a floor.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to rolling tools for flooring. In particular, the present invention relates to a floor rolling apparatus and associated methods.

BACKGROUND

(2) Laying floors is an arduous task that can be strenuous on a person's body. Oftentimes flooring such as vinyl must be rolled after it has been applied in order to ensure it is flat. Other rollers in the art do not leverage a person's arms and weight to ease the user's body strain.

(3) There exists a need in the art for a floor roller that leverages a person's own body and weight to facilitate the floor rolling process. Therefore, there exists a need in the art for a floor rolling apparatus as claimed and described herein.

SUMMARY OF THE INVENTION

(4) Embodiments of the present invention are related a floor rolling apparatus with a pair of

longitudinal support members, an upper lateral support connecting from one longitudinal support member to the other, a lateral grip connecting from one longitudinal support member to the other, a lower reinforcement assembly, and a roller assembly with removably attached roller. The longitudinal support members may include a respective pair of upper vaulted ends and the lower reinforcement assembly may include an L-bracket. The L-bracket may include an L-bracket bottom, an L-bracket side, and an angled support member. The angled support member may extend from the L-bracket bottom to the L-bracket side. Additionally, the roller assembly may include a roller brace structured to fit a removable roller therein. There may be a lower extension structured to fit the roller assembly between the pair of longitudinal support members and the roller assembly may include a roller with a pair of opposing compressible attachment knobs structured to removably attach the roller within a pair of knob holes on opposing sides of the L-bracket. Furthermore, the upper lateral support may be positioned on an upper portion of the apparatus and the lateral grip may be positioned medially on the apparatus.

(5) Another embodiment of the invention may include a floor rolling apparatus including a pair of longitudinal support members with a respective pair of upper vaulted ends, an upper lateral support connecting from one longitudinal support member to the other, a lateral grip connecting from one longitudinal support member to the other, a lower reinforcement assembly, and a roller assembly with removably attached roller. The longitudinal support members may include a first leveraging section and a second leveraging section that together with the upper lateral support may be structured to leverage a user's weight, lower arms and upper arms to facilitate rolling the apparatus roller on a floor.

(6) In this embodiment, the vaulted ends may be structured to rock along a user's upper arms when the roller is being used and may be angled on at least two sides of the longitudinal support members allowing the apparatus to be reversible. The longitudinal support members may include a fitting notch at one respective end allowing respective compressible attachment knobs to rest therewithin when the roller is attached. The longitudinal support members may be structured to be spaced apart by a length equivalent to an average user's shoulder width. A lower extension may be structured to fit the roller assembly between the pair of longitudinal support members. Additionally, the roller assembly may be attached to the lower reinforcement assembly via fasteners fitting through adjustable fitting holes within the roller assembly. The lower reinforcement assembly may include an L-bracket with an angled support member.

(7) Yet another embodiment of the invention may include a floor rolling apparatus with a pair of longitudinal support members including a respective pair of upper vaulted ends, an upper lateral support connecting from one longitudinal support member to the other, a medially positioned lateral grip connecting from one longitudinal support member to the other, a lower reinforcement assembly, a roller assembly with removably attached roller, and a lower extension structured to fit the roller assembly between the pair of longitudinal support members. The longitudinal support members may be structured to be spaced apart by a length equivalent to shoulder width and may include a first leveraging section and a second leveraging section that together with the upper lateral support may be structured to leverage a user's weight, lower arms and upper arms to facilitate rolling the apparatus roller on a floor. The lateral grip may be positioned medially on the apparatus and the upper lateral support may be positioned on an upper portion of the apparatus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of the floor rolling apparatus according to an embodiment of the invention.

(2) FIG. 2 is a front perspective view of the floor rolling apparatus according to an embodiment of

the invention illustrated in FIG. 1.

(3) FIG. 3 is a segmented top half view of the floor rolling apparatus illustrated in FIG. 1.

(4) FIG. 4 is a segmented bottom half view of the floor rolling apparatus illustrated in FIG. 1.

(5) FIG. 5 is a segmented bottom portion view of the floor rolling apparatus illustrated in FIG. 1.

(6) FIG. 6 is a segmented bottom portion view of the floor rolling apparatus illustrated in FIG. 1.

(7) FIG. 7 is a segmented bottom view of the floor rolling apparatus illustrated in FIG. 1 with detached roller.

(8) FIG. 8 is a demonstrative environmental front view of the floor rolling apparatus illustrated in FIG. 1.

(9) FIG. 9 is a demonstrative environmental front view of the floor rolling apparatus illustrated in FIG. 1.

(10) FIG. 10 is a demonstrative environmental side perspective view of the floor rolling apparatus illustrated in FIG. 1.

(11) FIG. 11 is a demonstrative environmental side perspective view of the floor rolling apparatus illustrated in FIG. 1.

DETAILED DESCRIPTION OF THE INVENTION

(12) The present invention will now be described in detail with reference to the accompanying drawings. The embodiment descriptions are illustrative and not intended to be limiting in any way. Other embodiments of the invention will readily suggest themselves to persons with ordinary skill in the art after having the benefit of this disclosure. Accordingly, the following embodiments are set forth without any loss of generality and without imposing limitation upon the claimed invention.

(13) Directional terms such as “above” “below” “upper” “lower” and other like terms are used for the convenience of the reader in reference to the drawings. Additionally, the description may contain terminology to convey position, orientation, and direction without departing from the principles of the present invention. Such positional language should be taken in context of the represented drawings.

(14) Quantitative terms such as “generally” “substantially” “mostly” and other like terms are used to mean that the referred object, characteristic, or quality constitutes a majority of the referenced subject. Likewise, use of the terms such as first and second do not necessarily designate a limitation of quantity. Such terms may be used as a method of describing the presence of at least one of the referenced elements or may provide a means of differentiating orientation. The meaning of any term within this description is dependent upon the context within which it is used, and the meaning may be expressly modified.

(15) Referring now to FIG. 1, the floor rolling apparatus **100**, hereinafter apparatus **100**, will be described more fully. The apparatus **100** may include a pair of side longitudinal supports **101** with upper vaulted ends **102** at one end. In some embodiments, the side longitudinal supports **101** may be spaced apart by a length equivalent to an average person's shoulder width.

(16) Positioned at a lower end may be a lower reinforcement assembly **103** structured to connect with and reinforce a roller assembly **108** that holds a roller **104**. The roller assembly **108** may be positioned between the pair of side longitudinal support members **101** at a lower end of the apparatus **100**. Further positioned between each side longitudinal support **101** and connecting from one side longitudinal support **101** to the other may be a lateral grip **105** and an upper lateral support **106**. The upper lateral support **106** may be positioned on an upper portion of the apparatus **100** and the lateral grip **105** may be positioned medially on the apparatus **100**. Each may be connected to the side longitudinal supports **101** via fasteners **107**, which may be nut and bolt combinations.

(17) FIGS. 2, 3 and 4 show how the apparatus **100** may be viewed as an upper segment **201** and a lower segment **202**. The upper segment **201** may range from the upper vaulted ends **102** to the lateral grip **105** and the lower segment **202** may range from the lateral grip **105** to the roller assembly **108**.

(18) Furthermore, the upper segment **201** may include a first leveraging section **211** and a second

leveraging section **212**. The first leveraging section may range from the upper vaulted ends **102** to the upper lateral support **106** and the second leveraging section **212** may range from the upper lateral support **106** to the lateral grip **105**.

(19) FIG. **3** shows how the vaulted ends **102** may include two angled sides culminating into a rounded vertex allowing the apparatus **100** to be reversible. As will be shown hereinafter, the vaulted ends **102** facilitate the apparatus' **100** use since the angled sides allow for the device to rock along a user's upper arm without having an end corner awkwardly thrusting into their upper arm.

(20) FIG. **4** takes a closer look at the lower segment **202**, which includes the lower reinforcement assembly **103** and the roller assembly **108**. The lower reinforcement assembly **103** may include a pair of L-brackets **401**. The L-brackets **401** may include L-bracket bottoms **402**, L-bracket sides **403**, and angled support members **404**. The L-bracket sides **403** may be affixed to an interior of each respective side longitudinal support **101**. The L-bracket bottoms **402** may provide a lateral support for the lower reinforcement assembly **103** to attach. Furthermore, the angled support members **404** may extend from the L-bracket bottoms **402** to the L-bracket sides **403** and may assist with distributing the weight and force applied to the apparatus **100** when in use as well as assist with maintaining the integrity of the lower segment **202** of the apparatus **100**.

(21) The roller assembly **108** may fit within a lower extension housing **425**. This may be an extension of the side longitudinal supports **101** structured to fit a roller brace **405** therein. The roller brace **405** may have a brace foundation **406** and a brace side **407**. Additionally, the roller **104** may removably attach at opposite ends to the brace sides **407** via compressible attachment knobs **408**.

(22) FIGS. **5** and **6** take a closer look at the lower segment **202**. As shown specifically in FIG. **5**, the brace foundation **406** may secure to the L-bracket bottoms **402** via brace fasteners **502**. In some embodiments, this may be a nut and bolt fastener. Furthermore, the brace sides **407** may be secured to the side longitudinal support **101** within the lower extension housing **425** via L-bracket fasteners **501**.

(23) FIG. **6** illustrates that the roller **104** may be removably detached from the L-bracket **402**. This is because the compressible attachment knob **408** may fit within a knob hole **601** within the brace side **407**. Furthermore, the compressible attachment knobs **408** in the attached state may bias within fitting notches **602** embedded within respective ends of the side longitudinal supports **101** after they have permeated the knob hole **601**. But, when the compressible attachment knob **408** is compressed and the roller **104** pulled, it may remove from the roller brace **405**.

(24) FIG. **7** shows a completely detached roller **104**, but illustrates where the compressible knobs **408** may fit within the knob holes **601** and fitting notches **602**. Furthermore, the brace fasteners **502** may fit within adjustable fitting holes **701** on the brace foundation **406**. So, the roller assembly **108** may be attached to the lower reinforcement assembly **103** via fasteners **502** fitting through adjustable fitting holes **701** within the roller assembly **108**. Also shown, is how the brace sides **407** extend past the respective ends of the side longitudinal supports **101** so that the compressible knobs **408** may rest within the fitting notches **602** of each end of the side longitudinal supports **101**.

(25) FIGS. **8** and **9** show a front environmental view of how the apparatus **100** may be used. Referring to FIG. **8**, a user **801** may grasp the lateral grip **105** with both hands **804** while the lateral support **106** is abutting perpendicularly the user's **801** upper arm **802**. The first leveraging section **211** may abut the user's upper arm **802** longitudinally. The roller **104** is on the floor **805** and the user **801** is in a generally upright stance.

(26) FIG. **9** shows the user **801** bent over using their body weight to extend the apparatus **901** outward rolling the roller **104** away from their body. The lateral support **106** along with the first leveraging section **211** press against the user's upper arms **802** while the user **801** is leaning into the lateral grip **105** to roll the apparatus **100** outward on the floor **104**. Therefore, the apparatus **100** is using bodyweight, leverage, and momentum to do the rolling work while the user **801** is simultaneously being supported by the apparatus **100**.

(27) FIGS. **10** and **11** are side environmental views of how the apparatus **100** may be used. FIG. **10**

shows again that the user **801** may grasp the lateral grip **105** with both hands **804** while the lateral support **106** is abutting perpendicularly the user's **801** upper arm **802**. The first leveraging section **211** may abut the user's upper arm **802** longitudinally. The roller **104** is on the floor **805** and the user **801** is in a generally upright stance. The second leveraging section is abutting the user's forearm **803**.

(28) FIG. **11** shows the user **801** bent over using their body weight to extend the apparatus **901** outward rolling the roller **104** away from their body. The lateral support **106** along with the first leveraging section **211** press against the user's upper arms **802** while the user **801** is leaning into the lateral grip **105** to roll the apparatus **100** outward on the floor **104**. The second leveraging section **212** is using the user's forearms **803** for added leverage. The apparatus **100** is using bodyweight, leverage, and momentum to do the rolling work while the user **801** is simultaneously being supported by the apparatus **100**. Stated differently, the longitudinal support members **101** may include a first leveraging section **211** and a second leveraging section **212** that together with the upper lateral support **106** and lateral grip **105** are structured to leverage a user's weight, lower arms **803** and upper arms **802** to facilitate rolling the apparatus roller **100** on the floor **805**.

(29) Furthermore, this angle shows that the vaulted ends **102** prevent the ends of the side longitudinal supports **101** from awkwardly thrusting into the user's **801** upper arms. They instead rock along a user's upper arms **802** when the roller **104** is being used. Additionally, the vaulted ends **102** may be angled on at least two sides of the longitudinal support members **101** allowing the apparatus **100** to be reversible.

Claims

1. A floor rolling apparatus comprising: a pair of parallel longitudinal support members with a respective pair of upper vaulted ends; a first leveraging section and a second leveraging section; an upper lateral support connecting from one longitudinal support member to the other; a lateral grip connecting from one longitudinal support member to the other; a lower reinforcement assembly; and a roller assembly with removably attached roller; wherein the lateral grip is perpendicular to the parallel longitudinal support members and configured as a user's hand grip; wherein the first leveraging section ranges from the upper vaulted ends to the upper lateral support and the second leveraging section ranges from the upper lateral support to the lateral grip; and wherein the parallel longitudinal support members are parallel from the roller assembly to the vaulted ends; wherein the roller is positioned between each respective parallel longitudinal support member; wherein the lateral grip is positioned below the upper lateral support and the first leveraging section; wherein the first leveraging section of each parallel longitudinal support member comprises a straight extension configured to leverage against a user's upper arm; and wherein the floor rolling apparatus is symmetric from the front, back, and sides and configured as reversible.
2. The floor rolling apparatus of claim 1 wherein opposing ends of the upper lateral support are terminated by the longitudinal support members and the longitudinal support members extend straight and distally from the lateral support members toward an upper end of the floor rolling apparatus before tapering into the vaulted ends.
3. The floor rolling apparatus of claim 1 wherein the lower reinforcement assembly comprises an L-bracket.
4. The floor rolling apparatus of claim 3 wherein the L-bracket includes an L-bracket bottom, an L-bracket side, and an angled support member.
5. The floor rolling apparatus of claim 4 wherein the angled support member extends from the L-bracket bottom to the L-bracket side.
6. The floor rolling apparatus of claim 1 wherein the roller assembly includes a roller brace configured to fit a removable roller therein.
7. The floor rolling apparatus of claim 1 further including a lower extension configured to fit the

roller assembly between the pair of longitudinal support members.

8. The floor rolling apparatus of claim 1 wherein the roller assembly comprises a roller with a pair of opposing compressible attachment knobs configured to removably attach the roller within a pair of knob holes on opposing sides of an L-bracket.

9. The floor rolling apparatus of claim 1 wherein the upper lateral support is positioned on an upper portion of the apparatus and the lateral grip is positioned medially on the apparatus.

10. A floor rolling apparatus comprising: a pair of parallel longitudinal support members with a respective pair of upper vaulted ends; an upper lateral support connecting from one longitudinal support member to the other; a lateral grip connecting from one longitudinal support member to the other; a lower reinforcement assembly; and a roller assembly with removably attached roller; wherein the upper lateral support is perpendicular to the parallel longitudinal support members and configured as a user's hand grip; wherein the parallel longitudinal support members are parallel from the roller assembly to the vaulted ends; wherein the roller is positioned between each respective parallel longitudinal support member; wherein the lateral grip is positioned below the upper lateral support and the first leveraging section; wherein the longitudinal support members include a first leveraging section and a second leveraging section that together with the upper lateral support are configured to leverage a user's weight, lower arms and upper arms to facilitate rolling the apparatus roller on a floor; and wherein the floor rolling apparatus is symmetric from the front, back, and sides and configured as reversible.

11. The floor rolling apparatus of claim 10 wherein the vaulted ends are configured to rock along a user's upper arms when the roller is being used.

12. The floor rolling apparatus of claim 10 wherein the vaulted ends are angled on at least two sides of the longitudinal support members allowing the apparatus to be reversible.

13. The floor rolling apparatus of claim 10 wherein the longitudinal support members include a fitting notch at one respective end allowing respective compressible attachment knobs to rest therewithin when the roller is attached.

14. The floor rolling apparatus of claim 10 wherein the longitudinal support members are configured to be spaced apart by a length equivalent to a user's shoulder width.

15. The floor rolling apparatus of claim 10 further including a lower extension configured to fit the roller assembly between the pair of longitudinal support members.

16. The floor rolling apparatus of claim 10 wherein the roller assembly is attached to the lower reinforcement assembly via fasteners fitting through adjustable fitting holes within the roller assembly.

17. The floor rolling apparatus of claim 10 wherein the lower reinforcement assembly includes an L-bracket with an angled support member.

18. A floor rolling apparatus comprising: a pair of parallel longitudinal support members with a respective pair of upper vaulted ends; an upper lateral support connecting from one longitudinal support member to the other; a medially positioned lateral grip connecting from one longitudinal support member to the other; a lower reinforcement assembly; and a roller assembly with removably attached roller; a lower extension configured to fit the roller assembly between the pair of longitudinal support members; wherein the upper lateral support is perpendicular to the parallel longitudinal support members and configured as a user's hand grip; wherein the parallel longitudinal support members are configured to be spaced apart by a length equivalent to shoulder width; wherein the parallel longitudinal support members are parallel from the roller assembly to the vaulted ends; wherein the roller is positioned between each respective parallel longitudinal support member; wherein the lateral grip is positioned below the upper lateral support and the first leveraging section; wherein the parallel longitudinal support members include a first leveraging section and a second leveraging section that together with the upper lateral support are configured to leverage a user's weight, lower arms and upper arms to facilitate rolling the apparatus roller on a floor; and wherein the floor rolling apparatus is symmetric from the front, back, and sides and

configured as reversible.

19. The floor rolling apparatus of claim 18 wherein opposing ends of the upper lateral support are terminated by the longitudinal support members and the longitudinal support members extend straight and distally from the lateral support members toward an upper end of the floor rolling apparatus before tapering into the vaulted ends.

20. The floor rolling apparatus of claim 18 wherein the upper lateral support is positioned on an upper portion of the apparatus.
